

Dealing with Stochastic Epidemiological Models - an Overview of Calibration Techniques

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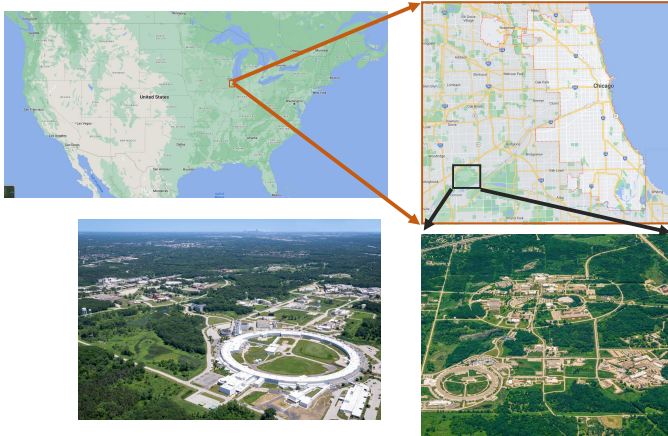


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- **Computer simulation**

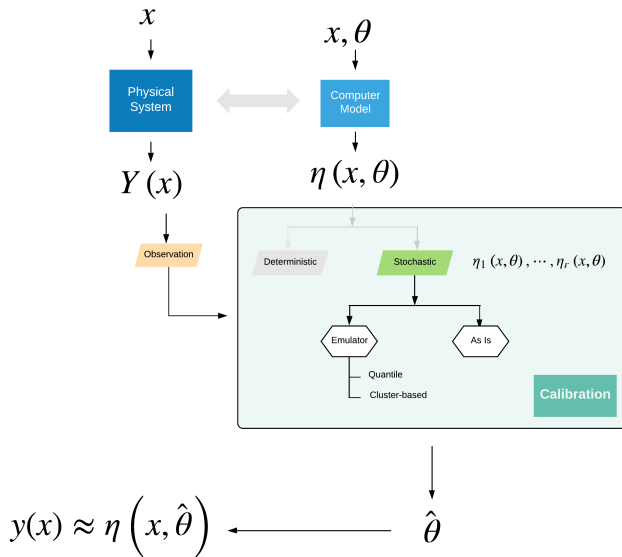
Want to study physical system, but can not directly work with it. Build a mathematical model which mimics the true system, and work with that.

- **Deterministic vs Stochastic**

Some models return same output when executed repeatedly at same input, and some return different output.

- **Statistical Inference**

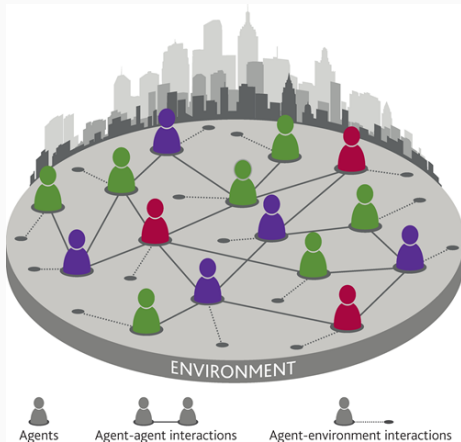
Need statistical tools to draw inference on the physical system based on computer model. Requires extra attention for stochastic simulations.



- **Motivating example** – Agent Based Model
- **Calibration via surrogates** –
 - Quantile based emulator
 - Common random number based emulator

Agent-Based Model

- Autonomous agents,
- Agent properties,
- Set of rules.



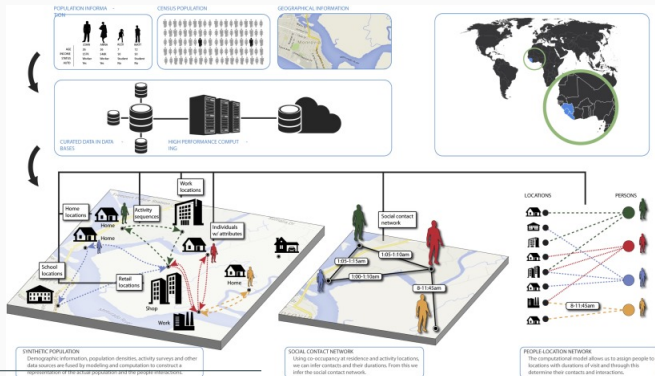
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¹<https://www.bankofengland.co.uk/quarterly-bulletin/2016/q4/agent-based-models-understanding-the-economy-from-the-bottom-up>

Agent-Based epidemic simulation

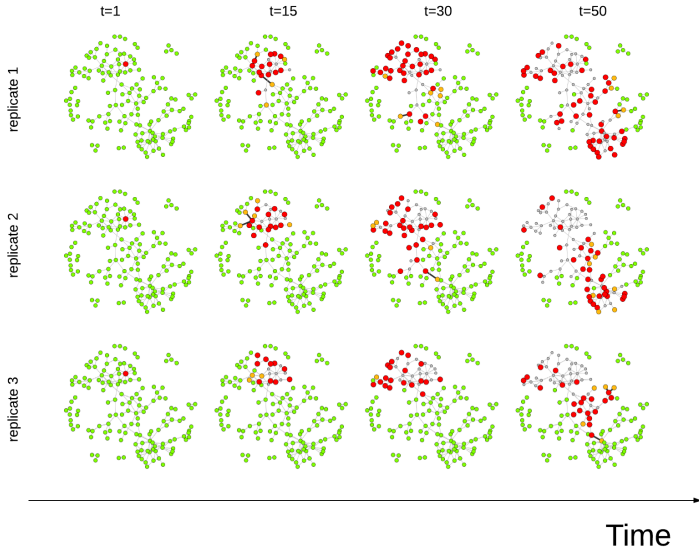
Ingredients:

- Synthetic Population
- Activity information
- Social contact network

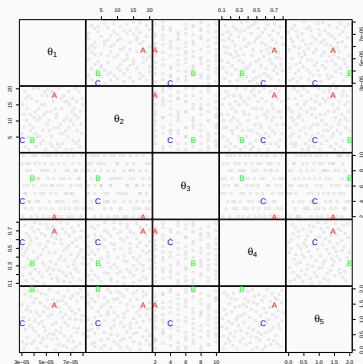


²image courtesy: Henning Mortveit

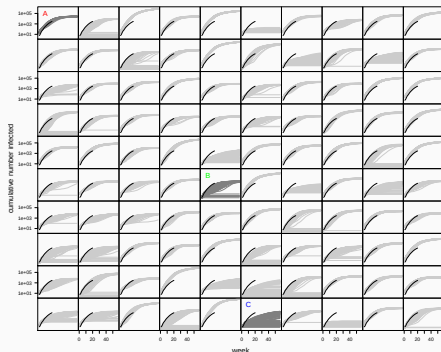
Agent-Based epidemic simulation (SEIR)



Statistical problem



Space-filling symmetric LH
design of size 100.



Simulation output at 100
design points, each with 100
replicates (in grey) along with
the actual observations (in
black).

- Fit a surrogate model $\eta(x, \theta)$ (using a Gaussian process)
 - Quantile GP
 - Mixture GP
 - Common random number GP
- Use $\eta(x, \theta)$ to search for $\hat{\theta}$ such that $Y(x) \simeq \eta(x, \hat{\theta})$.
 - Likelihood model:

$$Y(x) = \eta(x, \theta) + e \quad e \sim N(0, \sigma^2)$$

Gaussian Process - short overview

- $z(x) \sim GP(\mu(x), \Sigma(x))$.
- Observations: $z_1(x_1), \dots, z_n(x_n)$

$$\mathbf{z}_{n \times 1} \sim N(\boldsymbol{\mu}_{n \times 1}, \boldsymbol{\Sigma}_n), \quad \Sigma_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{\theta}\right)$$

- Prediction at new x^* : $z(x^*) | \mathbf{z}_{n \times 1} \sim N(\mu^*, \Sigma^*)$.

$$\mu^* = \boldsymbol{\Sigma}(x^*)^T \boldsymbol{\Sigma}_n^{-1} \mathbf{z},$$

$$\Sigma^* = \boldsymbol{\Sigma}_n - \boldsymbol{\Sigma}(x^*)^T \boldsymbol{\Sigma}_n^{-1} \boldsymbol{\Sigma}(x^*),$$

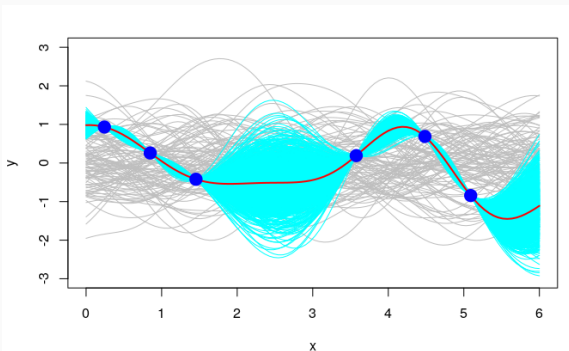
where, $\boldsymbol{\Sigma}(x^*) = (\boldsymbol{\Sigma}(x^*, x_1), \dots, \boldsymbol{\Sigma}(x^*, x_n))^T$.

Gaussian Process - short overview

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Challenges

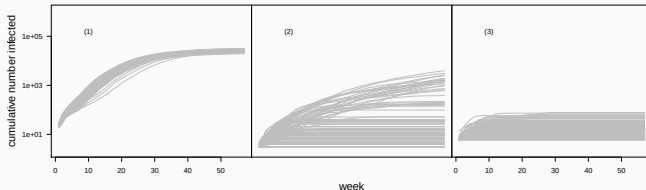


Figure 1: Simulation output at three different input settings, with replication

Notes:

- Clearly the replicate variability is not gaussian.
- Also not homoskedastic.
- Deal with multivariate response.

(1) Qunatile Emulation on univariate response

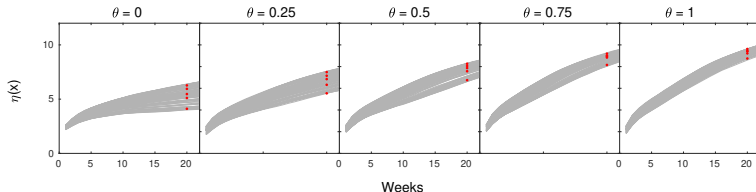


Figure 2: A collection of 100 epidemic ABM simulations at 5 different input settings. The red dots shows the empirical quantile estimates $Q(\alpha)$ of the logged epidemic size at week 20 with $\alpha = .05, .275, .5, .725, .95$.

Augment an additional input to $\eta(\theta)$ which is the quantile level α .

(1) Qunatile Emulation on 1-D response

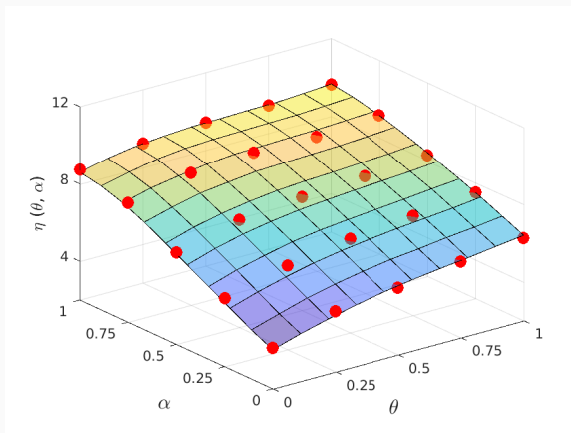
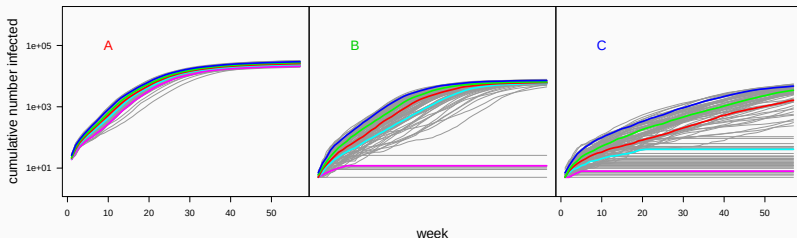


Figure 3: Posterior mean surface of the simulator output as a function of input θ and quantile α

(1) Quantile Emulation on multivariate response

Preprocess the data - empirical quantiles

- Pointwise $\alpha = (.05, .275, .5, .725, .95)$ -quantiles estimated from the 100 replicates.
- α -quantile estimate may not correspond to an actual simulation trajectory.

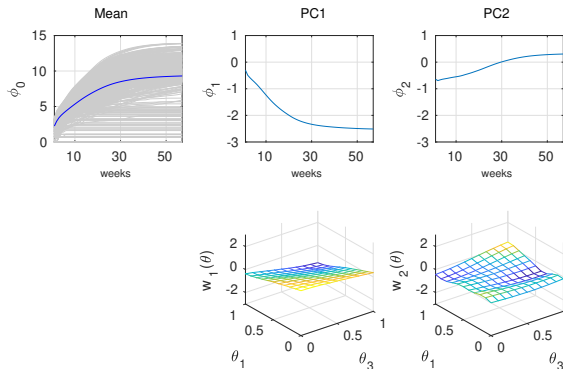


Simulated trajectories in grey, and estimated quantiles in color.

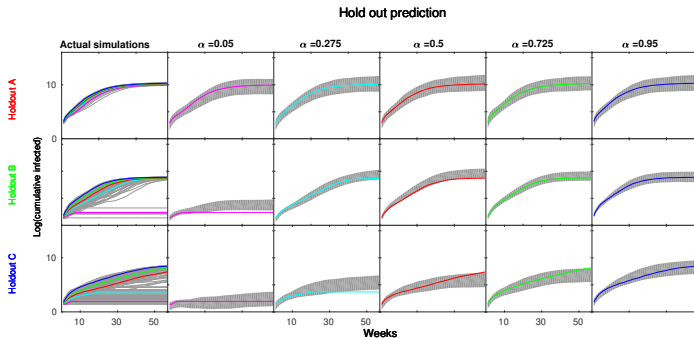
(1) Quantile Emulation on multivariate response

Basis representation of multivariate output

$$\eta(\boldsymbol{\theta}, \alpha) = \phi_0 + \sum_{k=1}^{p_\eta} \phi_k w_k(\boldsymbol{\theta}, \alpha) + \epsilon_{w0},$$

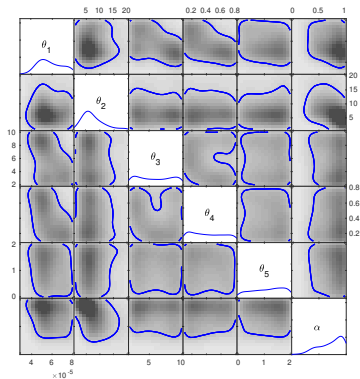


(1) Qunatile Emulation on multivariate response



The first column from the left shows the actual simulations and their empirical quantiles at the three design inputs denoted by their colors from the design. Next 5 columns show the pointwise 90% credible intervals of the posterior emulator prediction at those model inputs.

(1) Calibration using the quantile based emulator



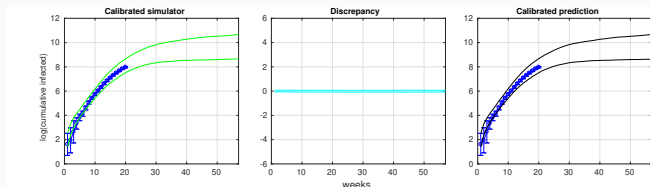
Full bayesian solution

Estimated posterior distribution of the parameters $(\theta_1, \dots, \theta_5, \alpha)$.

The diagonal shows the estimated marginal posterior pdf for each parameter; the off-diagonal images give estimates of bivariate marginals; to contour lines show estimated 90% hpd regions.

- Posterior is obtained from gaussian likelihood, and suitable priors for the parameters, e.g. gamma priors for error precisions, sparsity enforced prior for GP covariance parameters etc.

(1) Prediction using the quantile based emulator



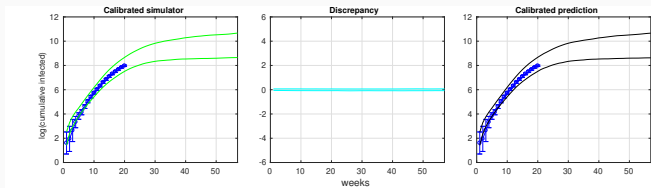
Decomposition of the posterior prediction for the epidemic curve.

Left: posterior 90% intervals for the calibrated emulator $\eta(\theta)$. The intervals are due to uncertainty regarding both θ and the GP emulator of ABM.

Middle: posterior 90% intervals for δ .

Right: posterior 90% intervals for the actual epidemic curve: $\eta(\theta) + \delta$. The blue dots show the observed counts, along with 1 sd bars given by the square root of $\text{diag}(\Sigma_y)$.

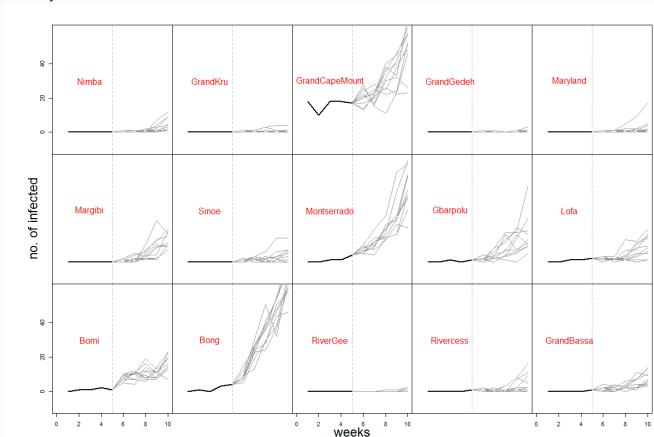
(1) What did we achieve and what we didn't



- **The good stuff:**
 - Reasonably accurate prediction.
 - Attached uncertainty.
 - Able to address non-Gaussian output distribution.
- **The not so good stuff:**
 - Not ideal for ongoing pandemic, i.e., online calibration.
 - . . .

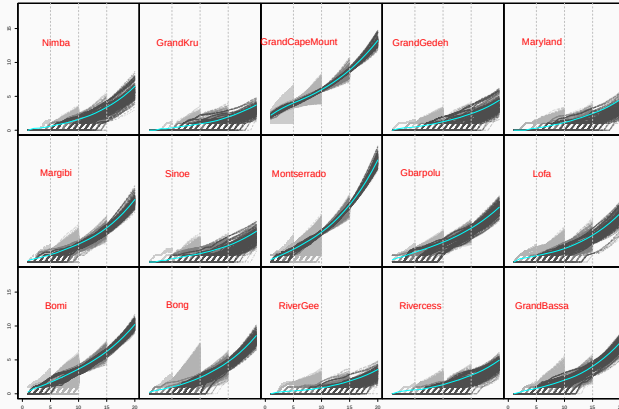
The primary concern

We want to move forward in time with good simulated curves (trajectories).



Basic Idea

Sequentially select plausible models from stochastic simulator.



The statistical challenge

- Our Quantile-GP surrogate **can not** reproduce individual replicate trajectory from the simulator. It can only guess the quantiles of the stochastic output at each θ .

$$\eta(\theta, \alpha) = \phi_0 + \sum_{k=1}^{p_\eta} \phi_k w_k(\theta, \alpha) + \epsilon_{w0}.$$

- Construct a surrogate for the stochastic simulation that can reproduce individual trajectory.

We can map each trajectory to random seed, can we use random seed as an input to the surrogate, i.e., $\eta(\theta, s)$?

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(2) Common Random Number Gaussian Process (CRNGP)

- Define a GP prior, that can explore similarities in random seed:

$$\eta(\boldsymbol{\theta}, s) \sim GP(0, \boldsymbol{\Sigma}(\boldsymbol{\theta}, s))$$

(what do we mean by similarity in random seed in the epidemic model context?)

- Treat time indices as another input to deal with multivariate output, i.e., define a surrogate $\eta(\boldsymbol{\theta}, s, t)$.
- We come up with a new covariance kernel. Given a $0 < \rho < 1$,

$$\boldsymbol{\Sigma}((\boldsymbol{\theta}_i, t), s), ((\boldsymbol{\theta}_j, t'), s')) = \begin{cases} \boldsymbol{\Sigma}((\boldsymbol{\theta}_i, t), (\boldsymbol{\theta}_j, t')) & \text{if } s = s' \\ \boldsymbol{\Sigma}((\boldsymbol{\theta}_i, t), (\boldsymbol{\theta}_j, t')) \times \rho & \text{o.w.} \end{cases}$$

(2) Common Random Number Gaussian Process (CRNGP)

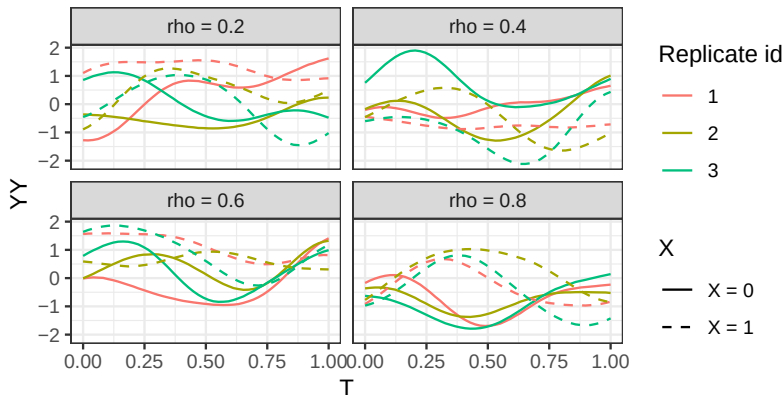


Figure 4: Realizations from CRNGP for different values of ρ (keeping the other GP hyperparameters fixed). Each line corresponds to a random realization from the CRNGP which is vector output (trajectory) indexed by variable T (scaled to $[0, 1]$).

(2) Common Random Number Gaussian Process (CRNGP)

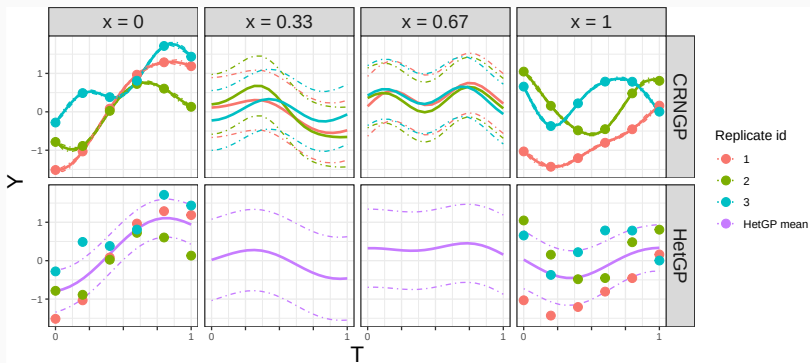


Figure 5: Comparison between fitted CRNGP (top) and Heteroskedastic GP (bottom) on vector-valued response at 4 different values of input x . Each color corresponds to a random seed value except the purple which represents the predicted mean surface from the HetGP model.

(2) Common Random Number Gaussian Process (CRNGP)

Calibration as optimization

Goal:

$$\epsilon(\boldsymbol{\theta}, s) = \sum_t (\eta_t(\boldsymbol{\theta}, s) - y_t^{obs})^2$$

$$(\hat{\boldsymbol{\theta}}, \hat{s}) = \arg \min_{\boldsymbol{\theta} \in \Theta, s \in \mathcal{S}} \epsilon(\boldsymbol{\theta}, s)$$

, $\mathcal{S}$ is finite.

We use *Thompson sampling* for finding $(\hat{\boldsymbol{\theta}}, \hat{s})$.

Thompson Sampling

Steps:

- **Initialization** Fit a GP surrogate on an initial set of simulations.
- **Repeat**
 - Draw random sample from fitted GP (as opposed to the mean).
 - Find the $(\boldsymbol{\theta}, s)$ where $\varepsilon(\boldsymbol{\theta}, s)$ is minimum.
 - Run a new simulation at $(\boldsymbol{\theta}, s)$ found on previous step.
 - Update GP surrogate.

Thompson Sampling

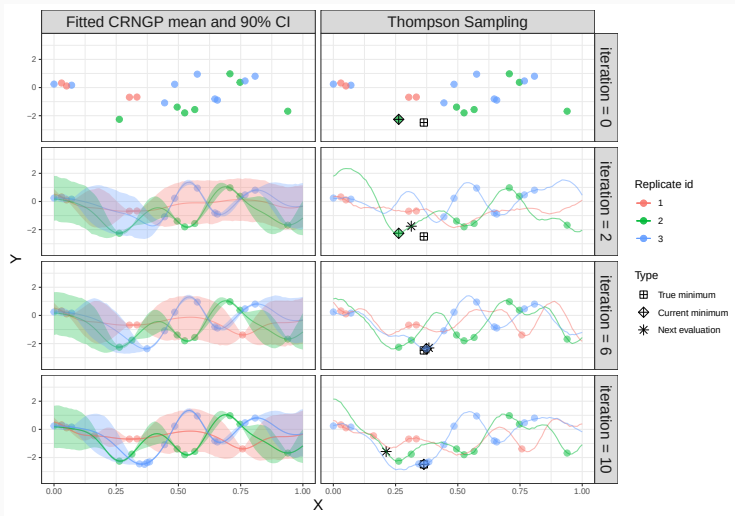


Figure 6: Thompson sampling based optimization using CRNGP surrogate.

(2) Common Random Number Gaussian Process (CRNGP)

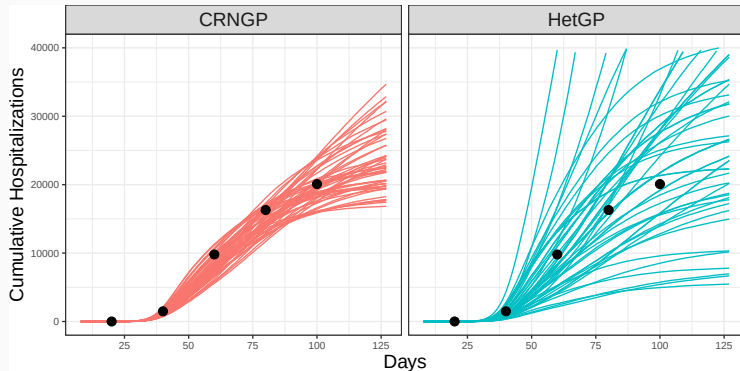


Figure 7: Best 50 trajectories of cumulative hospitalizations that are consistent with the observations denoted by black dots when using CRNGP and HetGP surrogates in Thompson sampling algorithm.

- CRNGP has clear advantage over Quantile GP surrogate, but still needs work to be done, e.g., finding out how many unique random seeds per parameter setting is sufficient.
- Stochastic computer models need attention, emulating mean and variance may not suffice.
- Sequential calibration is more relevant than one-shot calibration and new methods need to accommodate that.

- Fadikar, A., Higdon, D., Chen, J., Lewis, B., Venkatramanan, S., & Marathe, M. (2018). **Calibrating a stochastic, agent-based model using quantile-based emulation.** SIAM/ASA Journal on Uncertainty Quantification, 6(4), 1685-1706.
- Fadikar, A., Binois, M., Collier, N., Stevens, A., Toh, K. B., & Ozik, J. (2023). **Trajectory-oriented optimization of stochastic epidemiological models.** arXiv preprint arXiv:2305.03926. (accepted at Winter Simulation Conference 2023)